blood takes place in the ventricles. This type of circulation is called **incomplete double circulation**. The left atrium receives oxygenated blood and the right atrium receives deoxygenated blood. Pulmonary and systemic circuits are seen in Amphibians and Reptiles. The Crocodiles, Birds and Mammals have two auricles or atrial chambers and two ventricles, the auricles and ventricles are separated by inter auricular septum and inter ventricular septum. Hence there is complete separation of oxygenated blood from the deoxygenated blood. Pulmonary and systemic circuits are evident. This type of circulation is called c**omplete double circulation**.

7.4 Human circulatory system

The structure of the heart was described by **Raymond de viessens**, in 1706. Human heart is made of special type of muscle called the **cardiac muscle**. It is situated in the thoracic cavity and its apex portion is slightly tilted towards left. It weighs about 300g in an adult. The size of our heart is roughly equal to a closed fist. Heart is divided into four chambers, upper two small auricles or atrium and lower two large ventricles. The walls of the ventricles are thicker than the auricles due to the presence of papillary muscles. The heart wall is made up of three layers, the outer **epicardium**, middle **myocardium** and inner **endocardium**. The space present between the membranes is called pericardial space and is filled with pericardial fluid.

The two auricles are separated by **inter auricular septum** and the two ventricles are separated by **inter ventricular septum** The separation of chambers avoids mixing of oxygenated and deoxygenated blood. The auricle communicates with the ventricle through an opening called **auriculo ventricular aperture** which is guarded by the **auriculo ventricular valves**. The opening between the right atrium and the right ventricle is guarded by the **tricuspid valve** (three flaps or cusps), whereas a **bicuspid** (two flaps or cusps) or **mitral valve** guards the opening between the left atrium and left ventricle (Figure 7.6). The valves of the heart allows the blood to flow only in one direction, i.e., from the atria to the ventricles and from the ventricles to the pulmonary artery or the aorta. These valves prevent backward flow of blood.

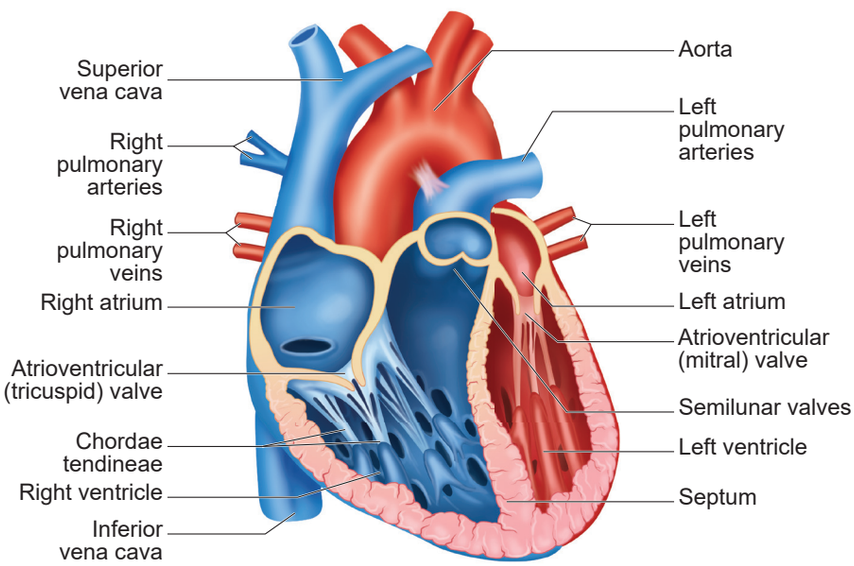


Figure 7.6 L.S of Heart

The opening of right and left ventricles into the pulmonary artery and aorta are guarded by aortic and pulmonary valves and are called **semilunar valves**.

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